

Transfer Learning with Fine-Tuning for Lithium-Ion Battery State-of-Health Estimation Across the BatteryLife Datasets: Interim Report

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Abstract—Estimating lithium-ion battery State-of-Health (SOH) from operational data is well established within a single dataset but degrades sharply once a trained model is moved to a battery dataset of different chemistry, cycling protocol, feature set, or prediction horizon. This interim report investigates whether transfer learning can reduce this gap and is structured around an embedded conference paper, with additional sections providing further context and analysis. Two questions are posed: whether fine-tuning a pre-trained LSTM network improves cross-dataset SOH accuracy over training from scratch (RQ1), and which factors — fine-tuning strategy, target-data fraction, feature overlap, and horizon alignment — determine when it does (RQ2). The conference paper pre-trains an LSTM on a source dataset and adapts it to each target under zero-shot inference, head-only fine-tuning and full fine-tuning across several data fractions, evaluating all 20 source–target pairs drawn from five BatteryLife datasets (140 transfer experiments) against from-scratch baselines and a computational-cost study. Its findings are that direct zero-shot reuse is never reliable (a 51%–3242% rise in RMSE); that fine-tuning restores and often improves on the baseline, the data-scarce target gaining most at roughly +20% from every source; that targets separate into source-independent and source-dependent classes; that the benefit switches on at a discrete target-data threshold rather than rising smoothly; and that a horizon mismatch between pre-training and deployment is a failure mode that adaptation cannot recover. The report concludes that the agreed project scope is met and on the strength of these results, specifies a second-half extension that broadens the benchmark from five to ten datasets and attaches multi-seed uncertainty bounds to every reported figure. The significance is a quantified map of where LSTM transfer learning can be trusted across battery datasets; and where it cannot, giving actionable guidance for reliable SOH-model deployment.

Index Terms—state of health, lithium-ion batteries, LSTM, transfer learning, fine-tuning, BatteryLife, cross-dataset generalisation, interim report

Word count: approximately 9300 words, excluding the abstract, nomenclature and the reference list.

NOMENCLATURE

Q_k Discharge capacity at cycle k .
 Q_{rated} Rated nominal capacity.
SOH(k) State of health at cycle k .

w Input window size (cycles).
 h Forecast horizon (cycles).
 d Number of input features.
 $\mathbf{x}_t$ Feature vector at cycle t .
 f_θ LSTM model with parameters θ .
 $\mathbf{h}_t$ LSTM hidden state at cycle t .
 $\mathbf{c}_t$ LSTM cell state at cycle t .
 R^2 Coefficient of determination.

ACRONYMS

BMS Battery Management System
CNX Constants, Noise, Variables (design framework)
EIS Electrochemical Impedance Spectroscopy
EOL End of Life
FC Fully Connected (layer)
LCO Lithium Cobalt Oxide
LFP Lithium Iron Phosphate
LSTM Long Short-Term Memory
MAE Mean Absolute Error
MSE Mean Squared Error
NCA Lithium Nickel Cobalt Aluminium Oxide
NMC Lithium Nickel Manganese Cobalt Oxide
PINN Physics-Informed Neural Network
RMSE Root Mean Squared Error
RUL Remaining Useful Life
SOH State of Health
V2B Vehicle-to-Building
V2G Vehicle-to-Grid

I. PREFACE: INTRODUCTION TO THE CORE PAPER

A. Status of the project and purpose of this report

This project investigates whether transfer learning with fine-tuning can improve lithium-ion battery SOH estimation across different datasets. The originally agreed deliverables included a from-scratch LSTM baseline, a pre-trained-and-fine-tuned transfer model, cross-dataset validation, a repeatable research package and an optional comparative analysis of fine-tuning strategies and data efficiency have all been completed and are

evidenced by the work presented here. The base scope of the project, as set out in the signed deliverables agreement, is met. The work that this report frames second semester is therefore an extension beyond the agreed scope rather than a completion of it, and the planning in Section XVIII is written on that basis.

Embedded within this report is a self-contained conference paper that consolidates the methodology, the full all-pairs cross-dataset experiment and the principal findings. The paper was submitted on 19 May 2026 to the 8th International Conference on System Reliability and Safety Engineering (SRSE 2026), under its Reliability Modelling and Analysis category and has since been accepted for presentation and publication. It was written to that venue’s strict page limit, and in compressing the work to that constraint a substantial amount of material needed for a complete account of the work had to be cut: the formal research questions, the experimental-design framework, the per-dataset baseline narratives, the cross-dataset computational-cost synthesis and an extended treatment of verification and limitations. This interim report restores that material.

B. Research questions

The project is focused around two research questions that the conference paper addresses but states only implicitly:

RQ1 (primary): Can transfer learning using a pre-trained LSTM model improve lithium-ion battery SOH prediction accuracy across different battery datasets compared to training from scratch?

RQ2 (secondary): Under what conditions does fine-tuning outperform training from scratch and which factors; fine-tuning strategy, target-data fraction, feature overlap and prediction-horizon alignment will most strongly determine transfer effectiveness?

The results generally support RQ1, although the effectiveness of transfer learning depends strongly on the target dataset. RQ2 became the main focus of the study, particularly in identifying when transfer learning succeeds or fails across datasets: the answer is a set of reliability boundaries; a two-regime split between source-independent and source-dependent targets, a step-function activation of transfer benefit at a discrete target-data threshold and prediction-horizon mismatch as a general failure mode that fine-tuning cannot repair.

C. Material restored from the core paper

The following sections provide additional detail on the experimental design, baseline analysis, computational cost study, transfer results, and limitations.

D. Statement of individual contribution

All experimental design, data preprocessing, model implementation, experiment execution, analysis, figure preparation and writing presented in the core paper and in this report are my own work. The co-authors listed on the embedded paper and on this report contributed in supervisory and review capacities only: Zongjun Li supervised the project for its first half and Huadong Mo has since taken over as supervisor; Runpu Wang reviewed drafts and made formatting and

phrasing suggestions; any other listed authors hold advisory roles. No co-author performed any of the modelling, coding, or experimental analysis. This is noted explicitly so that the technical work in the embedded paper is correctly attributed.

E. Report structure

Section I-D is followed by the core paper, reproduced without change. The sections after the conference paper (Sections III–XVI) provide the restored clarifications and Section XVIII sets out the second-half (extension) plan, including the experimental scope, the computational strategy, the risk treatment, and a weekly schedule with a Gantt chart.

F. Notation and conventions

A few conventions apply throughout, in both the embedded paper and the clarifications that follow it. SOH is normalised to each cell’s initial discharge capacity, with end-of-life taken at 80%. RMSE and MAE are computed on this normalised scale for every dataset except RWTH, which is reported in absolute ampere-hours; RWTH error magnitudes are not directly comparable with those of the other datasets. Transfer outcomes are expressed as the percentage change in RMSE relative to the from-scratch baseline, with positive values denoting improvement and negative values degradation. A source–target pair refers to a model pre-trained on the source dataset and adapted to the target and the three adaptation strategies are referred to as zero-shot (no target training), head-only fine-tuning (LSTM frozen, fully connected head retrained) and full fine-tuning (all parameters updated).

EMBEDDED CORE PAPER

The following paper is the central research artifact of this report and is reproduced exactly as written. The clarifications in the sections that follow it appear under my own name and do not alter any of its content; any corrections or extensions are made only in those later sections.

Failure Modes and Reliability Boundaries of Cross-Dataset Transfer Learning for Lithium-Ion Battery SOH Estimations

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Abstract—Reliable State-of-Health (SOH) estimation is essential for lithium-ion batteries deployed in electrified transport, grid-integrated storage, and Vehicle-to-Grid/Vehicle-to-Building applications. Although data-driven SOH estimators can achieve high accuracy within individual datasets, their reliability under cross-dataset deployment is limited by shifts in chemistry, operating protocol, feature availability, and prediction horizon. This paper investigates the failure modes and reliability boundaries of cross-dataset transfer learning for Long Short-Term Memory-based SOH estimation. Five public BatteryLife datasets are evaluated through 20 source-target combinations using zero-shot transfer, head-only fine-tuning, and full fine-tuning at multiple target-data fractions, resulting in 140 main transfer experiments. Results show that zero-shot transfer fails in all evaluated pairs, with RMSE degradation ranging from 51% to 3242%, indicating that direct deployment without target-domain adaptation is unreliable. Fine-tuning improves performance under identifiable conditions: the data-scarce CALCE target obtains approximately 20% RMSE improvement from all non-CALCE sources, whereas MATR consistently resists transfer. The evaluated targets further exhibit an empirical two-regime pattern, with some targets dominated by intrinsic target properties and others sensitive to source selection. A finer-grained fraction sweep reveals a threshold-like activation pattern, and prediction-horizon mismatch appears as a recurring cross-dataset failure mode. These findings provide reliability-oriented guidance for source selection, target-data collection, and safe deployment of transfer learning in battery SOH monitoring.

Index Terms—Lithium-Ion Batteries, State of Health, Transfer Learning, Cross-Dataset Generalisation, System Reliability, Battery Safety

I. INTRODUCTION

Lithium-ion batteries (LIBs) are increasingly deployed in electric vehicles (EVs), grid-scale storage, and building energy systems. As Vehicle-to-Grid (V2G) and Vehicle-to-Building (V2B) technologies enable EV batteries to supply energy back

to grids and buildings [1], inaccurate degradation assessment may lead to unsafe discharge depth, unexpected capacity loss, or reduced power availability [2]. Accurate real-time State-of-Health (SOH) monitoring is therefore essential for battery safety and system reliability.

In energy storage applications, machine learning has emerged as a promising approach for estimating battery state of charge, state of health, and remaining useful life [3]. However, data-driven models require substantial training data to achieve reliable performance, even though they avoid explicit electrochemical modelling [4]. Their reliability often degrades under cross-dataset deployment, as battery ageing trajectories vary across chemistries, manufacturers, cycling protocols, operating conditions, and available feature sets [5]. This dataset-specificity remains a critical barrier to safe and scalable BMS deployment.

Transfer learning offers a principled route for improving cross-dataset SOH estimation by transferring knowledge from a data-rich source domain to a data-limited target domain, thereby reducing the need to train a target-domain model from scratch [6]. Despite its potential, the reliability of transfer learning under broad cross-dataset deployment remains insufficiently characterised, particularly when source and target datasets differ in chemistry, feature availability, dataset scale, and prediction horizon [7]. These limitations make it difficult to derive practical guidance for deploying SOH models across heterogeneous battery datasets.

This paper investigates the failure modes and reliability boundaries of cross-dataset transfer learning for LSTM-based LIB SOH estimation. The main contributions are summarised as follows:

- 1) A reliability-oriented all-pairs evaluation across five

Unified Framework for Cross-Dataset SOH Prediction via Transfer Learning

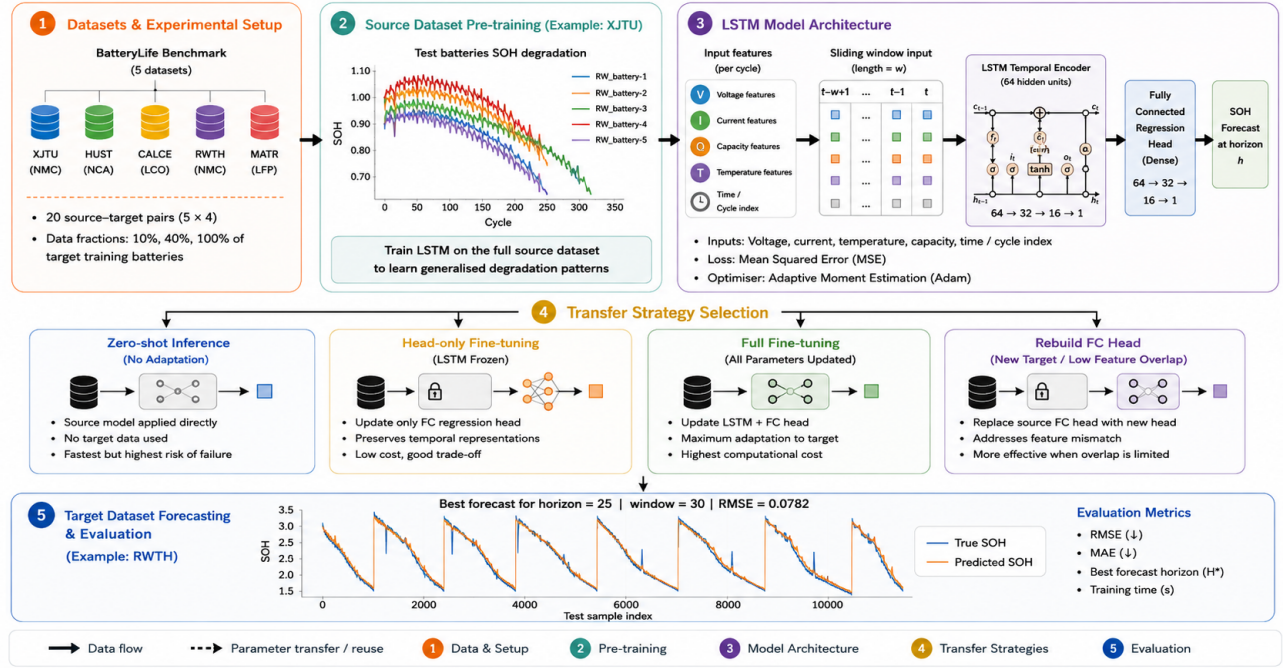


Fig. 1. Overview of the proposed cross-dataset SOH prediction framework. The workflow includes: (1) BatteryLife dataset setup across five datasets; (2) source-dataset pre-training using degradation trajectories; (3) LSTM model architecture with sliding-window inputs and an FC regression head; (4) transfer strategy selection, including zero-shot inference, head-only fine-tuning, full fine-tuning, and FC-head rebuilding; and (5) target-domain SOH forecasting and evaluation.

BatteryLife-derived datasets, covering 20 source-target pairs and 140 transfer experiments with zero-shot, head-only fine-tuning, and full fine-tuning strategies.

- 2) An empirical characterisation of transfer reliability boundaries, including source-dependent and source-independent target behaviours and a threshold-like activation pattern under limited target data.
- 3) An analysis of key cross-dataset failure modes, including prediction-horizon mismatch and feature-content mismatch, with guidance for source selection and safer SOH model deployment.

II. RELATED WORK

A. Battery Degradation and Reliability

Battery SOH reduction arises from chemistry-specific electrochemical ageing mechanisms [8]. Solid Electrolyte Interface (SEI) growth at the graphite anode consumes lithium inventory; lithium plating under high-rate charging creates dendrite-related safety risks. At the cathode, NMC cells suffer particle cracking; LFP cells exhibit lithium ordering phenomena; LCO cells are susceptible to structural collapse [1]. These mechanisms produce fundamentally different degradation trajectories across datasets, making cross-dataset reliability modelling inherently challenging. Data-driven methods learn degradation patterns directly from cycling data [9], but require large, dataset-specific training sets.

B. LSTM-Based SOH Estimation

Zhang et al. [10] demonstrated LSTM for battery remaining useful life prediction, showing the gating mechanism retains long-range temporal dependencies across hundreds of cycles. Wang et al. [11] conducted a comparative study of battery SOH prediction models, exploring Transformer-based approaches with reversible instance normalisation as an alternative to LSTM for sequence-based SOH estimation. Huang et al. [12] proposed a combined health indicator framework with LSTM, achieving RMSE below 1.44% across multiple datasets, and introduced fine-tuning and rebuilding transfer strategies that directly motivate this work. Lu et al. [13] demonstrated SOH estimation from partial charging curves without additional degradation experiments. Li et al. [14] showed that feature fusion with a hybrid CNN-LSTM architecture improves robustness under varying operating conditions.

C. Transfer Learning for Battery SOH

Ren et al. [5] demonstrated LSTM-based transfer learning reduces target-domain data requirements, characterising transfer benefit as a continuous function of data fraction. Qi et al. [7] extended transfer to EIS-based features for laboratory-to-field adaptation. Yao et al. [15] explored semi-supervised adversarial transfer for battery capacity estimation. The key gap across these studies is the absence of systematic all-pairs evaluation across diverse chemistries at scale — preventing characterisation of whether transfer effectiveness is driven by

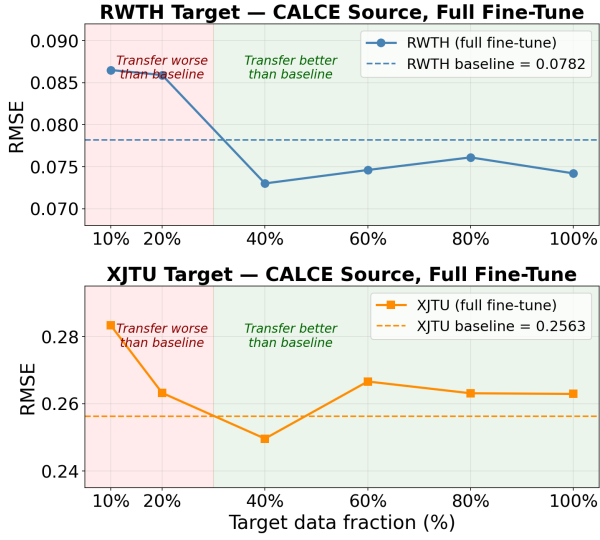


Fig. 2. Transfer activation threshold — CALCE source, full fine-tuning. RMSE compared to target data fraction for RWTH (top) and XJTU (bottom). Dashed lines show from-scratch baselines. Both targets cross below baseline at 40%; for XJTU, RMSE returns above baseline beyond 40% due to feature-space mismatch between CALCE and XJTU degradation dynamics.

source or target dataset properties, and whether failure modes such as horizon mismatch are dataset-specific or general.

III. METHODOLOGY

A. SOH Definition and Datasets

SOH is defined as the ratio of current to rated discharge capacity:

$$\text{SOH}(k) = \frac{Q_k}{Q_{\text{rated}}} \times 100\% \quad (1)$$

where Q_k is the discharge capacity at cycle k . A battery is considered end-of-life when SOH falls below 80% [8]. In V2G/V2B contexts, accurate SOH estimation near this threshold is safety-critical — an overestimate risks unsafe discharging of a degraded cell into a building load.

Five publicly available datasets are used, selected to maximise diversity in chemistry, feature richness, and operating conditions. Table I summarises their properties. CALCE is severely data-scarce (six LCO batteries, three features). MATR has the richest feature set (18 signals including DC internal resistance, energy efficiency, and charge throughput) encoding LFP fast-charge degradation dynamics absent from all other datasets. RWTH is the largest by battery count (668 cells) but reports capacity in Ah rather than normalised SOH. Datasets are sourced from the BatteryLife benchmark [16], which provides a standardised preprocessing and evaluation format for cross-dataset battery research.

B. LSTM Architecture and Training

The LSTM model consists of a single LSTM layer (hidden size 64), a dropout layer (rate 0.20), and three fully connected (FC) layers [64, 32, 16] producing a scalar SOH output —

TABLE I
SUMMARY OF DATASETS

Dataset	Cells	Features	Max cycles	Chemistry
XJTU	55	15	~350	NMC
HUST	77	13	~2000	LFP
CALCE	6	3	~1100	LCO
RWTH	668	15	~160	Mixed
MATR	139	18	~2000	LFP

27,521 trainable parameters, as illustrated in Panel 3 of Fig. 1. The cell state and hidden state are updated via:

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c) \quad (2)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad (3)$$

where $\mathbf{f}_t$, $\mathbf{i}_t$, $\mathbf{o}_t$ are the forget, input, and output gate activations. Input sequences use a sliding window: $\hat{y}_{t+h} = f_\theta(\mathbf{x}_{t-w+1}, \dots, \mathbf{x}_t)$ where w is the window size and h is the forecast horizon. Training uses Adam (lr=0.001), MSE loss, and early stopping (patience=8). Each dataset is split 70/15/15 by battery ID — preventing leakage and ensuring reliability metrics reflect genuine generalisation to unseen batteries.

C. Transfer Learning Strategies

All 20 source–target pair combinations are evaluated under three transfer settings, as illustrated in Panel 4 of Fig. 1. Zero-shot transfer applies the pretrained source model directly to the target test set without adaptation, representing the fastest but most failure-prone deployment case. Head-only fine-tuning freezes the LSTM backbone and retrain only the fully connected regression layers, preserving source-domain temporal representations while adapting the output mapping. Full fine-tuning updates all parameters on target data, providing the greatest adaptation capacity but also the highest risk of overfitting or catastrophic forgetting in low-data target regimes.

Both fine-tuning strategies are evaluated at 10%, 40%, and 100% of target training batteries in the main 20-pair sweep (7 configurations per pair, 140 total). A finer-grained fraction sweep at 10%, 20%, 40%, 60%, 80%, and 100% is additionally conducted using CALCE as source to precisely characterise the transfer activation threshold. Feature alignment uses the intersection of shared features between source and target. When fewer than five features are shared, the LSTM backbone is randomly reinitialised and only FC head weights are seeded from the source model.

IV. RESULTS

A. Baseline Performance

A systematic baseline sweep is conducted over window size $w \in \{10, 20, 30, 40, 50\}$ cycles and forecast horizon $h \in \{1, 10, 25, 50\}$ cycles, producing 100 configurations per dataset. The window size w controls how many historical cycles the LSTM receives as input, while the forecast horizon h determines how many cycles ahead the model predicts SOH. All 20 window-horizon combinations are evaluated on the

held-out test split and the optimal configuration per dataset — that achieving minimum test RMSE — is fixed as the source pre-training configuration for all subsequent transfer learning experiments involving that dataset. Table II reports the best RMSE per dataset. Fig. 3 shows RMSE heatmaps for RWTH and MATR, confirming that no single configuration generalises across datasets. HUST and MATR achieve the lowest RMSE (0.0095 and 0.0085), reflecting large clean training sets. XJTU achieves the highest RMSE (0.2563) from high cycle-to-cycle variability. A computational cost study confirms performance saturates before the full training set: for HUST, 40% data achieves RMSE=0.0100 at 447 s versus RMSE=0.0126 at 608 s for full training — a 42% reduction in training time with improved accuracy.

TABLE II
BEST BASELINE LSTM RESULTS PER DATASET

Dataset	W	H	RMSE	MAE
XJTU	30	1	0.2563	0.1802
HUST	30	10	0.0095	0.0073
CALCE	20	25	0.1998	0.0718
RWTH	30	25	0.0782	0.0602
MATR	50	10	0.0085	0.0065

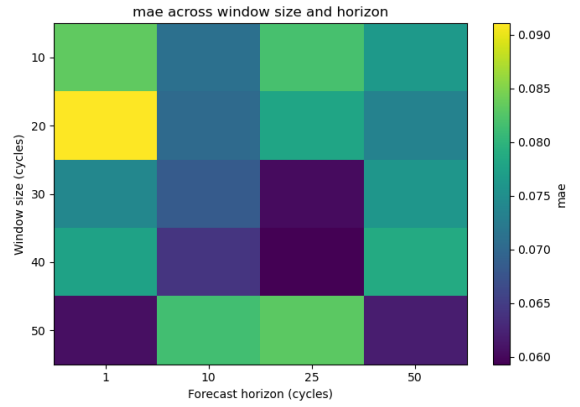
B. Transfer Learning Results

Table III summarises the best RMSE improvement over the from-scratch baseline for all 20 source-target pairs. Zero-shot transfer fails universally — RMSE degrades by 51%–3242% across all 20 pairs, with zero-shot R^2 reaching as low as -11.99 on RWTH, confirming predictions are actively worse than a simple mean predictor.

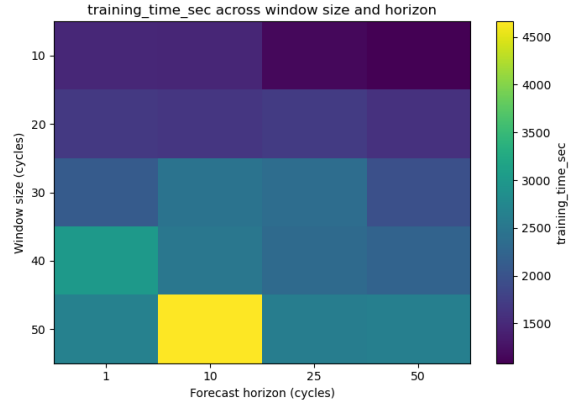
Fine-tuning consistently recovers and frequently exceeds baseline performance. CALCE achieves +20% improvement from all five source datasets, ranging from +20.6% (XJTU) to +21.7% (RWTH) — a narrow band across sources spanning NMC, NCA, LCO, LFP, and mixed chemistries. Fig. 4 shows per-source heatmaps for HUST and MATR as sources, confirming the CALCE improvement and MATR resistance patterns hold regardless of source. Table IV provides detailed RMSE results for CALCE as target. The RWTH full fine-tune at 10% diverges to RMSE=0.6974 due to prediction horizon mismatch (source H=25, target H=10) combined with very limited target data — a result that foreshadows Finding 4.

C. Transfer Benefit Activation and Data Threshold Analysis

A finer-grained fraction sweep with CALCE as source (Table V) reveals that for both RWTH and XJTU as targets, RMSE remains above baseline at 10% and 20% data fractions, then drops abruptly below baseline at exactly 40% — 187 batteries for RWTH (+6.6%) and 16 batteries for XJTU (+2.6%) — with negligible further improvement at 60%, 80%, or 100%. This step-function profile is structurally distinct from the continuous improvement curves reported by Ren et al. [5]. HUST and MATR fail to beat baseline at any fraction, for reasons discussed in Findings 4 and 2 respectively.



(a) RWTH



(b) MATR

Fig. 3. RMSE heatmaps (window size compared to forecast horizon) for RWTH and MATR baseline sweeps. RWTH shows uniform low error across configurations; MATR requires window size of 50 cycles for reliable prediction.

TABLE III
BEST TRANSFER LEARNING RMSE IMPROVEMENT OVER BASELINE (%)

Target Source	CALCE	HUST	RWTH	MATR
XJTU	+20.6	+6.2	+1.2	−218
HUST	+21.2	—	+0.7	−240
RWTH	+21.7	−8.0	—	−253
CALCE	+5.6	−17.9	+6.6	−336
MATR	+21.3	−8.2	+6.9	—

Positive values indicate improvement; negative values indicate degradation. — indicates the same source and target dataset or a setting not evaluated.

D. Computational Cost Compared to Accuracy

Table VI reports training time alongside RMSE for all strategies on RWTH (XJTU source) — the largest target dataset and the one where cost differences are most pronounced. Head-only fine-tuning at 40% achieves near-baseline accuracy (RMSE=0.0900) in 525 s. Full fine-tuning at 40% is the only configuration to beat baseline (+1.2%, RMSE=0.0773) but at 1007 s — nearly double the cost. Full fine-tuning at 100% costs 3089 s while performing worse than full fine-tuning at 40%, confirming that additional data beyond the activation threshold provides no accuracy benefit

TABLE IV
CALCE TARGET: RMSE BY SOURCE, STRATEGY, AND DATA FRACTION

Source	Strategy	10%	40%	100%
XJTU	Head-only	0.2487	0.1692	0.1619
	Full fine-tune	0.2419	0.1706	0.1587
HUST	Head-only	0.2417	0.1697	0.1582
	Full fine-tune	0.2547	0.1747	0.1574
RWTH	Head-only	0.3969	0.1741	0.1620
	Full fine-tune	0.6974	0.1621	0.1565
MATR	Head-only	0.2392	0.1630	0.1589
	Full fine-tune	0.2394	0.1630	0.1573
Baseline (from scratch)		0.1998		
Zero-shot (best source)		0.3015		

TABLE V
CALCE SOURCE: FULL FINE-TUNE RMSE COMPARED TO DATA FRACTION

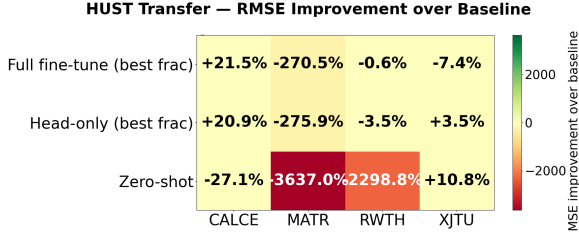
Target	10%	20%	40%	60%	80%	100%
RWTH	0.0865	0.0859	0.0730	0.0746	0.0761	0.0742
XJTU	0.2833	0.2632	0.2496	0.2666	0.2631	0.2629

Baseline: RWTH=0.0782, XJTU=0.2563. Bold: best per target. HUST and MATR fail to beat baseline at all fractions.

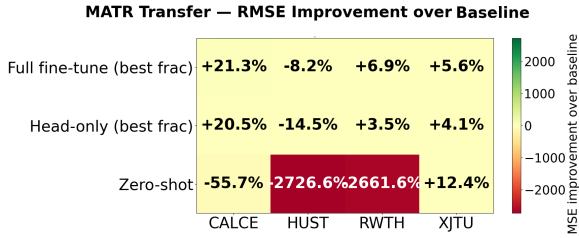
TABLE VI
RWTH TARGET (XJTU SOURCE): RMSE AND TRAINING TIME BY STRATEGY AND FRACTION

Strategy	Frac.	RMSE	Time (s)	compared to baseline
Zero-shot	—	1.8738	≈0	−2296%
Head-only	10%	0.0965	483	−23.4%
	40%	0.0900	525	−15.1%
	100%	0.0821	848	−5.0%
Full fine-tune	10%	0.0931	420	−19.1%
	40%	0.0773	1007	+1.2%
	100%	0.0833	3089	−6.6%
Baseline (from scratch)		396		0.0%

Bold: only configuration beating baseline. Baseline RMSE=0.0782.



(a) HUST



(b) MATR

Fig. 4. Transfer improvement heatmaps for HUST and MATR as sources. Both replicate the XJTU pattern: CALCE always improves (≈+21%), MATR always resists, confirming these are properties of the target dataset rather than the source.

at substantially higher computational cost.

V. DISCUSSION

A. Zero-Shot Deployment is a Reliability Risk

All 20 zero-shot evaluations produce catastrophic RMSE degradation. In V2G/V2B scenarios, a zero-shot model producing near-constant SOH estimates would fail to detect battery degradation entirely, creating direct safety risks for building electrical systems. The severity of failure correlates with domain shift — RWTH and MATR are the most resistant targets because their degradation dynamics are qualitatively unlike any source dataset — confirming that chemistry and protocol diversity compound the reliability risk.

B. Two-Regime Structure of Transfer Effectiveness

Transfer targets divide into two distinct regimes. *Source-independent targets* — CALCE and MATR — show improvement values essentially fixed regardless of source. CALCE achieves approximately +20% from every source (range: +20.6% to +21.7%); MATR resists transfer from every source

(no source achieves better than −218%). In both cases the outcome is determined by intrinsic target properties — data scarcity for CALCE, electrochemical uniqueness for MATR.

Source-dependent targets — HUST and RWTH — vary substantially with source choice. HUST ranges from −17.9% (CALCE) to +6.2% (XJTU), a swing of 24 percentage points; RWTH ranges from +0.7% (HUST) to +6.9% (MATR). For these targets, source selection is a meaningful engineering decision: the right source delivers reliable improvement, while the wrong source actively degrades performance. Notably, MATR and HUST share LFP chemistry yet MATR consistently resists transfer from HUST, confirming that chemistry alone does not determine transfer outcome — MATR’s fast-charge protocol encodes degradation dynamics absent from HUST’s multi-stage discharge cycling. This two-regime framing immediately tells a practitioner whether source selection matters for their target — something prior work evaluating only 2–4 pairs cannot provide.

C. Transfer Activates as a Step Function

The finer-grained CALCE sweep (Table V and Fig. 2) reveals that transfer effectiveness does not improve smoothly with additional data. For RWTH, RMSE remains above the from-scratch baseline at 10% (0.0865) and 20% (0.0859), then drops abruptly below baseline at 40% (0.0730, +6.6% improvement), with negligible further change at 60%, 80%, and 100%. The threshold crossing is clearly visible in the upper panel of Fig. 2 — the step-function profile is structurally distinct from the continuous improvement curves reported by Ren et al. [5], where transfer benefit accumulates gradually with additional target data.

For XJTU, shown in the lower panel of Fig. 2, the same threshold activation is observed at 40% (0.2496, +2.6% im-

provement). However, RMSE returns marginally above baseline at 60%, 80%, and 100% fractions. This behaviour arises because CALCE’s three smooth LCO features cannot capture XJTU’s high cycle-to-cycle variability — additional CALCE batteries reinforce representations that remain misaligned with XJTU’s noisy NMC degradation dynamics. This confirms that reaching the activation threshold is necessary but not sufficient: the quality of the source feature space ultimately constrains the ceiling of transferable benefit.

Practitioners should therefore concentrate data collection on reaching the activation threshold rather than pursuing larger datasets beyond it.

D. Horizon Mismatch is a General Failure Mode

Prediction horizon alignment between source pre-training and target deployment is an independent reliability constraint — and a *general* failure mode rather than a dataset-specific observation.

Initial evidence came from RWTH as source (pre-trained at $H=25$): applied to $H=10$ targets, RWTH produced uniquely severe zero-shot failures — RMSE=1.712 on HUST, 1.819 on CALCE, 2.049 on MATR — substantially worse than XJTU and HUST sources at $H=10$ despite comparable feature overlap.

Independent confirmation comes from CALCE as source (also pre-trained at $H=25$). Applied to HUST ($H=10$), CALCE produces zero-shot RMSE=0.1279 (−1240% degradation). Critically, fine-tuning does not resolve this: at no data fraction does CALCE beat the HUST baseline, with best result RMSE=0.0113 at 100% data (−17.9%), versus +6.2% achieved by XJTU at $H=10$. Fine-tuning cannot undo the representational incompatibility introduced by horizon mismatch during pre-training. With two independent sources — RWTH (mixed chemistry, 668 cells) and CALCE (LCO, 6 cells) — exhibiting the same failure mode, horizon alignment must be treated as a first-order constraint on source model selection, independent of feature overlap or chemistry.

E. Feature Content and Strategy

MATR shares 44% of features with RWTH and XJTU yet resists transfer from all sources, because its 10 unique signals encode fast-charge LFP degradation modes absent from all source training sets. Feature content — whether shared features encode the target’s dominant degradation mechanisms — is a more meaningful predictor than feature count. Table VI quantifies the strategy trade-off directly: head-only fine-tuning at 40% delivers near-optimal accuracy at roughly half the training time of full fine-tuning, and avoids the catastrophic forgetting risk that causes full fine-tuning to underperform head-only in low-data regimes.

VI. CONCLUSION

This paper investigated the failure modes and reliability boundaries of cross-dataset LSTM-based battery SOH transfer learning across 20 source-target pairs and 140 experiments.

Transfer targets divide into source-independent and source-dependent regimes, transfer benefit activates as a step function at a discrete battery count threshold, and prediction horizon mismatch is a general failure mode that fine-tuning cannot overcome. Zero-shot deployment is unsafe across all chemistry combinations tested. Future work will conduct multi-seed uncertainty quantification, investigate dataset similarity metrics as reliability predictors, and extend experiments to additional BatteryLife datasets.

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II. FROM THE CORE PAPER TO THE CLARIFICATIONS

One important point when interpreting the paper is that it was written primarily for a reliability and safety audience; the application of the work is towards Vehicle-to-Grid and Vehicle-to-Building deployment, where an over-optimistic SOH estimate can drive a degraded cell into an unsafe discharge. That register reflects the standard academic language of the target venue, but it maps directly onto this project's research questions: RQ1; the question of whether fine-tuning beats training from scratch, which the conference paper answers as the boundary between good or bad transfer and RQ2; the question of when, which the paper answers as its two-regime structure, activation threshold and horizon-mismatch failure mode. Both research questions are therefore addressing different aspects of the same underlying results.

The conference paper makes three contributions: a reliability-oriented all-pairs evaluation across five BatteryLife-derived datasets (20 source-target pairs and 140 transfer experiments); an empirical characterisation of transfer reliability boundaries, including source-dependent and source-independent target behaviours and a threshold-like activation under limited target data; and an analysis of the principal cross-dataset failure modes including prediction-horizon mismatch and feature-content mismatch, with guidance for source selection and safer deployment.

The remaining sections of this report expand five results from the conference paper in turn. First, zero-shot transfer fails in all 20 pairs, with RMSE degradation between 51% and 3242%. Second, fine-tuning recovers and frequently beats the from-scratch baseline, the data-scarce CALCE target improving by roughly 20% from every source. Third, targets fall into two regimes, source-independent (CALCE always improves, MATR always resists) and source-dependent (HUST and RWTH, where the choice of source swings the outcome). Fourth, transfer benefit activates abruptly at a discrete target-data threshold rather than accumulating smoothly. Fifth, a prediction-horizon mismatch between source pre-training and target deployment is a general failure mode, confirmed independently from two different source datasets, that fine-tuning cannot fix.

III. BACKGROUND

The conference paper provides only a brief background section, so additional context is included here.

A. The SOH estimation problem

Demand for lithium-ion batteries has grown rapidly with the electrification of transport and the expansion of renewable energy storage, making in-service monitoring of battery health a central engineering challenge. SOH is the standard ageing metric, defined as the ratio of present available discharge capacity to rated capacity, with a cell conventionally considered EOL once SOH falls below 80% [1]. SOH cannot be measured directly during normal operation; it must be inferred from measurable signals including terminal voltage, current and surface temperature. Which are derived quantities

such as differential voltage and incremental capacity, which renders estimation an inverse problem. The estimate is also consequential: battery management systems rely on SOH to set charging limits, power output, thermal management and cell balancing. Underestimating SOH can lead to premature battery replacement. Overestimating it is more serious because degraded cells may continue operating outside safe limits, with thermal-runaway consequences in safety-critical settings [2]. Accurate, deployable estimation is therefore important and the reliability boundaries of any cross-dataset estimator; which is the subject of the conference paper, warrant characterisation rather than assumption.

B. Battery degradation mechanisms

The data-driven modelling problem sits on top of real electrochemistry and the mechanisms differ by electrode and by chemistry. At the graphite anode the dominant ageing process is growth of the solid-electrolyte interphase (SEI), a passivating film whose continued growth consumes lithium inventory and electrolyte and so steadily reduces capacity; under low-temperature or high-rate charging, lithium plating can also occur, causing immediate capacity loss and longer-term dendrite-related safety risk [1]. At the cathode the mechanisms are chemistry-specific: NMC undergoes particle cracking and transition-metal dissolution at high state of charge; LFP is more structurally stable but exhibits lithium-ordering phenomena; LCO is susceptible to structural collapse at high charge voltage. Another complication is capacity recovery where rested cells can show a temporary capacity rebound through lithium redistribution and SEI relaxation, which produces non-monotonic SOH trajectories that complicate both modelling and feature engineering [2]. These mechanisms also predict which transfers should succeed. A target whose degradation is a smooth, predictable fade (LCO/CALCE) should accept almost any reasonable initialisation, while a target whose dynamics are dominated by fast-charge-specific modes encoded in unique features (LFP/MATR) should resist transfer from sources that never saw those modes. The empirical results match this prediction, which is one of the validation arguments in Section XII.

C. Families of SOH estimation methods

SOH estimation methods fall into three broad families [2]. Direct measurement methods can achieve high accuracy, although they usually require specialised equipment or controlled operating conditions. Coulomb counting needs a full known-condition cycle and EIS needs frequency-resolved stimulation. Model-based methods build explicit representations of the cell: equivalent-circuit models identified online via Kalman or particle filters, or higher-fidelity electrochemical models of lithium transport and reaction kinetics. Both require recalibration as the cell ages and the model diverges from measured behaviour, and their accuracy is ultimately bounded by model fidelity, which becomes problematic for novel chemistries [1]. Physics-informed neural networks (PINNs) are an emerging hybrid that embeds conservation laws into

the training loss, reducing data requirements and keeping predictions physically plausible; but PINNs require known governing equations for each chemistry, and deriving and validating a PINN for every chemistry in a cross-dataset study would be an overly time intensive endeavour. Data-driven methods learn degradation behaviour directly from cycling data and can be applied across different datasets. Their main drawback is the large amount of training data required. It is the data scarcity that motivates transfer learning. This project is data-driven and the PINN route is out of scope for the reason given above.

IV. LITERATURE REVIEW

The works that most directly shape this project span LSTM-based SOH estimation, the BatteryLife benchmark and transfer learning for battery SOH. The following studies are the most closely related to the project and help define the research gap addressed in this work.

A. Selection of the LSTM architecture

Among data-driven architectures, the LSTM is adopted deliberately rather than by default. Battery degradation is a sequential process unfolding over hundreds to thousands of cycles and the LSTM gating mechanism (forget, input and output gates) is designed to retain long-range temporal dependencies, which is what early work on RUL prediction demonstrated on NASA cells [3]. The architecture used here is closely related to the combined-health-indicator LSTM of Huang et al. [4], who also introduced the fine-tuning and rebuilding transfer strategies that this project generalises to an all-pairs design. The alternatives were considered and rejected: Transformer architectures need substantially more data and compute to train well, which is incompatible with CPU training and with target sets as small as one or two batteries; convolutional architectures capture local patterns but lack the explicit memory needed to carry degradation state across long sequences; and PINNs require the chemistry-specific formulations just discussed. LSTM is also the dominant architecture in the battery-SOH transfer-learning literature [4], [5], which keeps the results comparable with existing benchmarks.

B. The BatteryLife benchmark

The study is built on the BatteryLife benchmark [6], which aggregates cycling data from a large number of public experiments into a standardised format with consistent feature extraction, splits and metrics. This standardisation removes a major source of inconsistency in the battery-ML literature, where results are often reported on incompatible preprocessing pipelines and cannot be compared. Importantly for this work, BatteryLife was designed to support transfer-learning research: its diversity in chemistry, features, scale and operating conditions makes it a realistic platform for cross-dataset generalisation, and its own authors report poor zero-shot generalisation across constituent datasets. The observation that motivates the fine-tuning strategies studied here. Five constituent datasets are used in the current study; the extension intends to double this.

C. Transfer learning for SOH and the research gap

Transfer learning for battery SOH has attracted growing attention. Ren et al. [5] showed that LSTM-based transfer substantially reduces target-domain data requirements, with fine-tuned models beating from-scratch baselines on a fraction of the target data, but their work focused on within-chemistry transfer and characterised benefit as a smooth, continuous function of data fraction. Lu et al. [7] showed SOH can be estimated from partial charging curves without dedicated degradation experiments. Across this literature the consistent gap is the absence of a systematic all-pairs evaluation across a large, diverse benchmark spanning fundamentally different chemistries, feature sets and scales: most studies evaluate two to four pairs, usually within one chemistry family and do not characterise how transfer effectiveness depends on dataset properties. This is the gap the project addresses. Its contribution is: a systematic all-pairs cross-dataset evaluation across diverse BatteryLife datasets; a characterisation of the optimal fine-tuning data fraction across all pairs; evidence that prediction-horizon alignment is an independent, previously under-reported determinant of transfer effectiveness; and the CNX experimental-design framing that makes the controlled factors explicit. Notably, the activation of transfer benefit observed here is a step function at a discrete data threshold rather than the continuous curve reported by Ren et al. [5], which is a structural rather than incremental departure from the prior result.

V. RELATIONSHIP TO THE AGREED PROJECT DELIVERABLES

Because this report frames the second semesters work as an extension, the status of the contracted deliverables is established here. Table I maps each agreed deliverable to the evidence in the conference paper and the supporting code and results package, with a status against the original scope. Every contracted item, including the “optional” comparative analysis of fine-tuning strategies and data efficiency, has been delivered.

The cost study reported in the conference paper also serves here to establish that the from-scratch baselines are themselves already data-efficient, so the transfer improvements reported against those baselines are measured against a deliberately strict reference rather than an easily beaten one. This affects how the transfer results should be read: a +6.6% improvement on RWTH is an improvement over a baseline that has itself been driven close to its achievable floor.

VI. THE CNX EXPERIMENTAL-DESIGN FRAMEWORK

The experimental design was structured using a CNX (Constants, Noise, Variables) framework, shown in Fig. 1. Although omitted from the conference paper due to space constraints, the framework is important for explaining how the experiments were controlled.

Constants. The following were held fixed across every experiment: the LSTM architecture (single layer, hidden size 64, fully connected head [64-32-16-1], dropout 0.20, 27,521 trainable parameters); the train/validation/test split (70/15/15

TABLE I
AGREED DELIVERABLES MAPPED TO EVIDENCE AND STATUS.

Deliverable	Evidence	Status
Baseline model	From-scratch LSTM with full window-horizon sweep on five datasets; RMSE, MAE, training time and model size recorded (paper Table II; Sec. VIII).	Complete
Transfer learning model	Pre-trained LSTM with zero-shot, head-only and full fine-tuning at three target-data fractions (paper Sec. IV).	Complete
Cross-dataset validation	Complete all-pairs design: every dataset as source against the remaining four as targets, 20 pairs, 140 experiments (paper Table III).	Complete
Reproducible research package	Per-dataset notebooks, fixed-seed configuration, and saved forecasts/histories; preprocessing pipelines documented per dataset.	Complete
Optional comparative analysis	Strategy comparison (head-only vs. full), data-fraction sweeps, and a finer-grained activation-threshold sweep (paper Sec. IV-C, Table V).	Complete

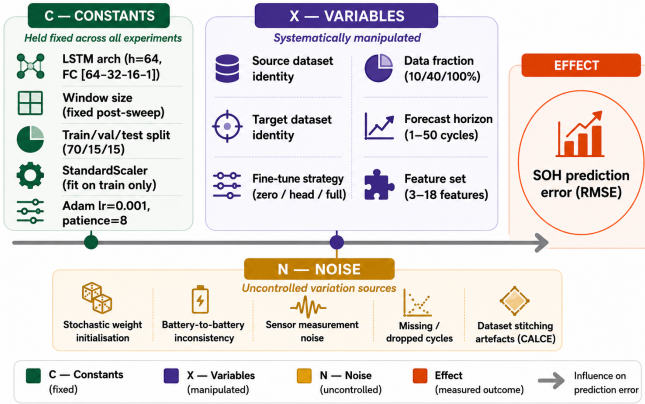


Fig. 1. CNX diagram classifying experimental factors. Constants (green) are fixed across all experiments; noise (yellow) are uncontrolled sources of variation; variables (blue) are the factors placed under systematic investigation; the effect (red) is SOH prediction RMSE.

by battery ID); feature normalisation (a StandardScaler fitted on the training split only); the optimiser (Adam, learning rate 10^{-3} , weight decay 10^{-5}); early stopping (patience 8, monitored on validation loss); the maximum epoch budget (120 for baselines, 40 for transfer); the random seed (42); and the loss function (MSE). Holding the architecture and optimiser constant holds the central comparison of the study: differences in outcome are attributable to the variables under investigation, not to incidental retuning.

Noise. The acknowledged uncontrolled sources of variation are: residual stochasticity in LSTM training despite the fixed seed; battery-to-battery inconsistency within a dataset arising from manufacturing tolerance and cell-to-cell variation; sensor measurement noise in the raw cycling data; missing or dropped cycles from acquisition failures or preprocessing exclusions; and the CALCE multi-file stitching artefacts discussed below. The single-seed limitation means these noise sources are not yet quantified — this is precisely the gap the extension’s multi-seed study is designed to close (Section XVIII).

Variables. The factors placed under investigation were the source dataset identity, the target dataset identity, the fine-tuning strategy (zero-shot, head-only, full), the fine-tuning data

fraction, the forecast horizon and the window size.

VII. EXTENDED METHODOLOGICAL CLARIFICATIONS

A. Data split and its interaction with transfer

Each dataset is split 70/15/15 by battery ID. Splitting by battery rather than by cycle ensures that no cell’s data appears in both training and test sets, which prevents the model from memorising individual degradation curves rather than learning generalisable degradation behaviour. For the transfer experiments this split structure is significant and is stated only briefly in the conference paper. The source model is pre-trained on the source dataset’s full training split. Fine-tuning then uses a fraction (10%, 40% or 100%) of the target’s training split only; the target’s validation and test splits are never seen during pre-training or fine-tuning. The target test split is held out entirely and used only for final evaluation. This ensures the reported transfer improvements reflect genuine generalisation to unseen target batteries rather than any form of leakage and that the zero-shot condition involves no target exposure of any kind: the source model meets the target test set without prior target exposure.

B. Feature-intersection alignment

Datasets differ in feature count (three for CALCE, up to eighteen for MATR), so a transfer between any pair must settle two different input spaces. The alignment rule is the intersection: the model is trained and fine-tuned using only the features present in both source and target. If a 15-feature source meets a 13-feature target with ten features in common, all transfer for that pair uses those ten shared features. This guarantees the model is never given, during fine-tuning, information that was unavailable during pre-training, and vice versa. When fewer than five features are shared, the LSTM backbone is randomly re-initialised and only the fully connected head weights are seeded from the source, because too little feature overlap makes the pre-trained temporal representations unlikely to be applicable. This rule has a direct consequence for the MATR results: because MATR shares only three features with CALCE and eight with the others, several MATR transfers operate in head-weights-only mode with a randomly initialised backbone, which limits how much pre-training can possibly contribute and is part of why MATR resists transfer.

TABLE II
BASELINE SUMMARY WITH COMPUTATIONAL-COST CONTEXT.

Dataset	Best RMSE	Train cells	Opt. frac.	Train (s)
XJTU	0.2563	38	60%	~400
HUST	0.0095	45	40%	447
CALCE	0.1998	4	40%	65
RWTH	0.0782	467	80%	396
MATR	0.0085	97	40%	~1300

Train cells: batteries in the full training split.

Opt. frac.: training fraction giving best cost-efficiency.

C. Reduction from a full-factorial design

With six variables under investigation, a full-factorial sweep would require $5 \times 4 \times 3 \times 3 \times 4 \times 5 = 3,600$ configurations, which is not achievable on the available CPU hardware. The design is reduced in a Taguchi-inspired manner. Window size and forecast horizon are treated as dataset-specific constants after the baseline sweep: the configuration that minimises test RMSE for each dataset is fixed and used as that dataset’s source pre-training configuration, reducing the active variable count from six to four for the transfer phase. The remaining four variables; source, target, strategy and data fraction are then fully crossed in the all-pairs design. The three data-fraction levels (10%, 40%, 100%) were chosen to sample the low-data, intermediate and saturated regimes of the learning curve; the cost study (Section IX) confirms three levels are sufficient to locate the optimum for each pair. This reduction concentrates effort on the interactions of greatest interest while controlling configuration-level noise through the pre-optimised window–horizon selection.

D. Implementation and tooling

The experimental code is implemented in Python. Model definition and training use PyTorch; data parsing and the per-cycle feature tables use pandas and NumPy; feature scaling and the evaluation metrics use scikit-learn; and figures are produced with Matplotlib and Plotly. AI coding assistants were used to support scripting, refactoring, and documentation, with all experimental design, configurations, results, and their interpretation defined and verified by the author. Two tooling-related limitations follow from this stack and are relevant to the results: all training was performed single-threaded on CPU, which bounded the feasible number of configurations and repeats, and library default behaviours (for example, non-deterministic CUDA kernels) are not a factor here only because no GPU was used, which is a point to be revisited if the extension moves to accelerated hardware.

VIII. PER-DATASET BASELINE ANALYSIS

The conference paper reports only the best RMSE per dataset. The per-dataset behaviour behind those numbers, which indicates whether the baselines are sound, is summarised here. Table II places the best baseline alongside the training-set size and the cost-efficient training fraction identified in Section IX.

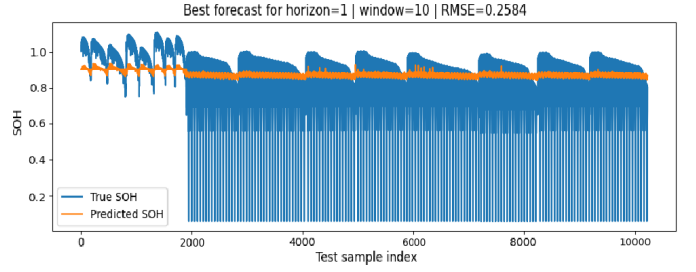


Fig. 2. XJTU best baseline forecast (horizon 1). The prediction follows the mean degradation trend but cannot reproduce the high-amplitude cycle-to-cycle oscillation characteristic of XJTU’s irregular cycling protocol which is the source of its high baseline RMSE.

XJTU is the weakest baseline ($RMSE = 0.2563$, $MAE = 0.1802$ at window 30, horizon 1). The weakness is a property of the data, not the model: XJTU’s six batches were cycled under variable rest times, temperature fluctuations and irregular protocols, which produces a high-amplitude, cycle-to-cycle oscillation in the true SOH signal. As Fig. 2 shows, the model learns a smoothed mean trajectory and cannot track that oscillation, and performance degrades steadily with horizon ($RMSE$ rises to roughly 0.35–0.59 at horizon 50). This baseline establishes a noise floor set by the dataset’s intrinsic variability.

HUST achieves $RMSE = 0.0095$ ($MAE = 0.0073$, $R^2 = 0.969$ at window 30, horizon 10), the second-best baseline. This reflects clean, smooth, consistently decreasing SOH trajectories; the model captures the long-term trend reliably and errors grow only modestly with horizon. The full HUST training split is sizeable (86,968 rows across 45 batteries) and longer window–horizon combinations reach roughly 1,600 s of training time.

CALCE reports $RMSE = 0.1998$ ($MAE = 0.0718$ at window 20, horizon 25), but this figure requires careful interpretation. The raw, unnormalised RMSE before stitching correction ranged from 8.92 to 14.0, an order of magnitude larger than every other dataset, because CALCE’s per-battery trajectory is reconstructed by stitching multiple Excel channel sheets and the stitching introduces SOH values spiking above 4.0 near cycle 250 and unexplained dropouts toward zero (Fig. 3). These were handled by clipping SOH to $[0, 1.2]$ and flagging affected cycles. After correction the normalised baseline is comparable to XJTU but is achieved with only six batteries and three features. This weak baseline, produced under severe data scarcity, is the condition under which transfer is expected to help; every source improves CALCE by roughly +20%.

RWTH achieves $RMSE = 0.0782$ ($MAE = 0.0602$ at window 30, horizon 25) across 10,706 test samples drawn from 668 batteries. This was the most statistically reliable baseline in the study. Its RMSE stays within 0.078–0.116 across all 20 window–horizon configurations, indicating low sensitivity to those hyperparameters. One caveat dominates its interpretation: RWTH reports capacity in absolute Ah rather than normalised SOH (the curves in Fig. 4 range from roughly

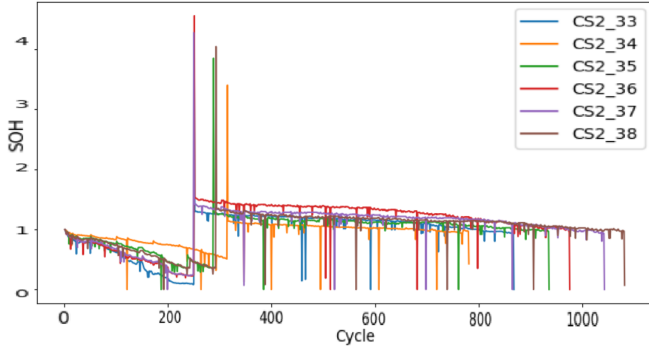


Fig. 3. CALCE SOH degradation curves. Stitching artefacts are visible near cycle 250 (SOH spikes above 4) and are handled in preprocessing by outlier clipping. With only six batteries, CALCE is the most data-scarce dataset in the study.

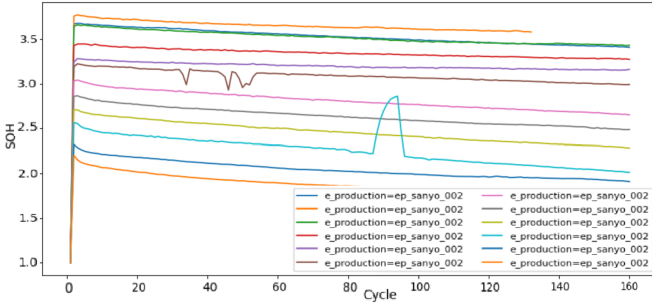


Fig. 4. RWTH SOH degradation curves (sample). Capacity is reported in absolute Ah rather than normalised SOH; cells degrade smoothly over roughly 160 cycles. The Ah scale means RWTH RMSE values are not directly comparable with the normalised datasets.

1.7 to 3.8 Ah), so its RMSE is not on the same scale as the normalised datasets and absolute comparisons across datasets are inappropriate. RWTH cells also degrade over only ~ 160 cycles, giving shallow temporal depth despite the large cell count.

MATR achieves the best baseline of all five ($RMSE = 0.0085$, $MAE = 0.0065$ at window 50, horizon 10) across 13,842 test samples. This reflects the richest feature set (18 signals including DC internal resistance, energy efficiency, charge throughput, energy throughput and a time-temperature integral) and a large, diverse training set (97 batteries, 83,971 rows). MATR is uniquely sensitive to window size with performance improving steadily from window 10 to window 50 at short horizons. This which suggests longer temporal context is especially valuable for fast-charge degradation. It is also the most expensive to train, reaching 4,662 s at the best configuration on CPU.

IX. COMPUTATIONAL-COST STUDY: CROSS-DATASET SYNTHESIS

The conference paper highlights the HUST and RWTH cost curves; the consistent pattern across all five datasets is stated in full here because it underpins both the baseline-efficiency argument and the data-fraction choices in the transfer design.

In every dataset, predictive performance saturates well before the full training set is used and the full set offers no measurable accuracy benefit at substantially higher cost.

For HUST, 40% of the training data gives the best short-to-medium-horizon result ($RMSE = 0.0100$ at 447 s versus 0.0126 at 608 s for the full set), a 42% reduction in training time with improved accuracy. For RWTH, performance is near-flat from 40% onward for horizons 1, 10 and 25, with 80% giving the best horizon-1 result ($RMSE = 0.0753$). CALCE shows a sharp threshold rather than a smooth curve: a single training battery (fractions 0.1 and 0.2) gives $RMSE \approx 0.25$ –0.29, while two batteries (fraction 0.4) drops it to about 0.146, essentially matching the full-data result at half the cost. MATR is non-monotonic, intermediate fractions (40% for most horizons) beat the full set, because the final 20% of MATR’s training batteries are noisier and skew learned signal. The cross-dataset reading is that additional target data is not a reliable to achieve accuracy, consistent with the transfer fraction sweep and supporting the credibility of the activation-threshold finding in the conference paper.

The cost-accuracy trade-off this implies is shown directly in Fig. 5, which plots training time against RMSE for every strategy and data fraction with XJTU as source. The zero-shot point sits at essentially zero cost but unusable error; head-only fine-tuning at 40% consistently sits within the lower-left, Pareto-optimal region, delivering near-best accuracy at a fraction of the cost of full fine-tuning at 100% data. This is the basis for the conference paper’s recommendation of head-only at 40% as a sensible default. The per-dataset learning curves in Fig. 6 make the saturation explicit: for both RWTH and HUST, RMSE flattens by 40–60% of the training data across short-to-medium horizons, so the final 40–60% of the data yields little additional accuracy at two-to-three times the training time. Taken together, the Pareto view and the learning curves establish that the cost of the from-scratch baselines is already near-minimal, which is the strict reference the transfer improvements are measured against.

X. TRANSFER LEARNING: PER-SOURCE ANALYSIS

The conference paper presents the transfer results primarily by target. The complementary per-source view clarifies why some sources are broadly useful and others are not.

XJTU is among the strongest sources, achieving positive transfer on three of four targets: CALCE +20.6%, HUST +6.2% and RWTH +1.2%, failing only on MATR (−218%). Its HUST result is the single best transfer outcome in the study in relative terms, which is consistent with XJTU and HUST sharing similar slow/mixed-protocol degradation behaviour.

HUST mirrors XJTU closely as a source: CALCE +21.2%, RWTH +0.7%, XJTU +2.6%, and MATR −240%. The near-identical CALCE figure from a different source is the first sign that the CALCE benefit is a property of the target, not the source.

RWTH is a weaker source than its size would suggest: CALCE +21.7% and XJTU +2.7%, but HUST −8.0% and MATR −253%. RWTH’s zero-shot failures are severe across

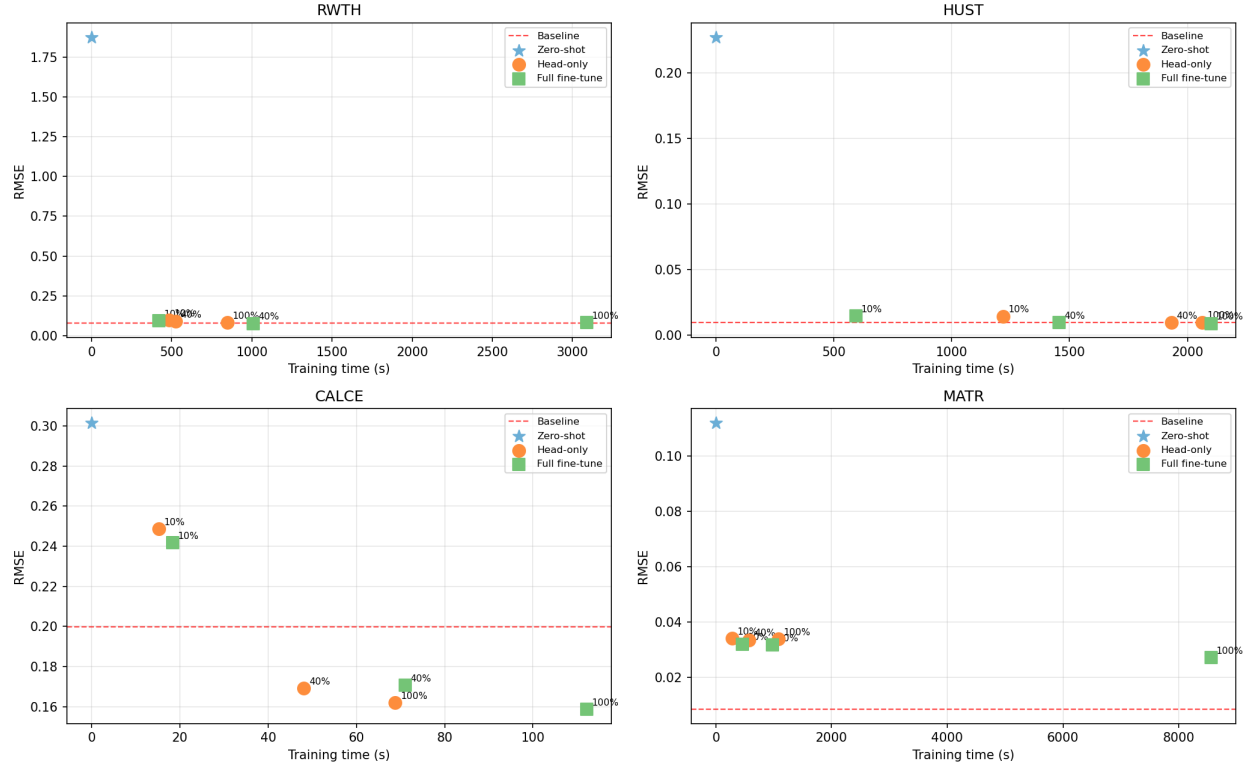


Fig. 5. Cost-error trade-off (XJTU source). Stars: zero-shot; circles: head-only; squares: full fine-tune, with data fractions labelled. Lower-left is better. Head-only at 40% sits on the Pareto front for CALCE, HUST and RWTH, achieving near-best accuracy at a fraction of the training cost of full fine-tuning at 100%.

all targets, which is the clearest single piece of evidence for the horizon-mismatch failure mode. RWTH was pre-trained at horizon 25 while the targets use horizon 10.

CALCE, the most data-scarce source (three features, six batteries, always operating in head-weights-only mode), is the weakest source overall but still beats the baseline on two targets: RWTH +6.6% and XJTU +2.6%, while failing on HUST (−17.9%) and MATR (−336%). That a three-feature source can still improve a 668-battery target is a useful lower-bound result.

MATR is an unexpectedly effective source: despite its specialised feature set and large feature mismatch with every target, it beats the baseline on three of four: XJTU +5.6%, RWTH +6.9% (the best RWTH result of any source) and CALCE +21.3%, failing only on HUST (−8.2%). MATR’s rich temporal representations show a broadly useful initialisation even when only the shared discharge-capacity, voltage and current features.

Across sources, neither dataset size nor feature richness alone predicts transfer effectiveness: RWTH is large but a poor source for HUST and MATR is feature-rich but fails on HUST while excelling on RWTH. What predicts effectiveness is the alignment between source and target degradation dynamics. This forms a hypothesis the extension’s dataset-similarity analysis is intended to test directly.

XI. ADDITIONAL TRANSFER DIAGNOSTICS (XJTU SOURCE)

The conference paper reports the transfer outcomes compactly through improvement tables and per-source heatmaps for HUST and MATR. Using XJTU; one of the two strongest sources, as a representative case shows the strategy-level structure behind those summary numbers and the same pattern reoccurs for the other sources.

Fig. 9 (Annex A) shows the RMSE of all seven transfer configurations (zero-shot, plus head-only and full fine-tuning at 10%, 40% and 100%) for each target, against the from-scratch baseline. The figure shows the project’s central result directly: the zero-shot bar towers above the baseline on every target, while the fine-tuned bars collapse to near or below it on CALCE, HUST and RWTH. MATR is the exception, every fine-tuned bar remains above the baseline. Fig. 7 condenses the same information into an improvement heatmap over the baseline: the zero-shot row is uniformly and severely negative, the fine-tuning rows turn positive for CALCE (+20.6%), HUST (+6.2%) and RWTH (+1.2%) and the MATR column stays negative (−218%) regardless of strategy or fraction. These two views are consistent with the all-pairs improvement table in the embedded paper, and they show that the principal per-target numbers are not driven by a single favourable configuration but hold across the strategy and fraction grid.

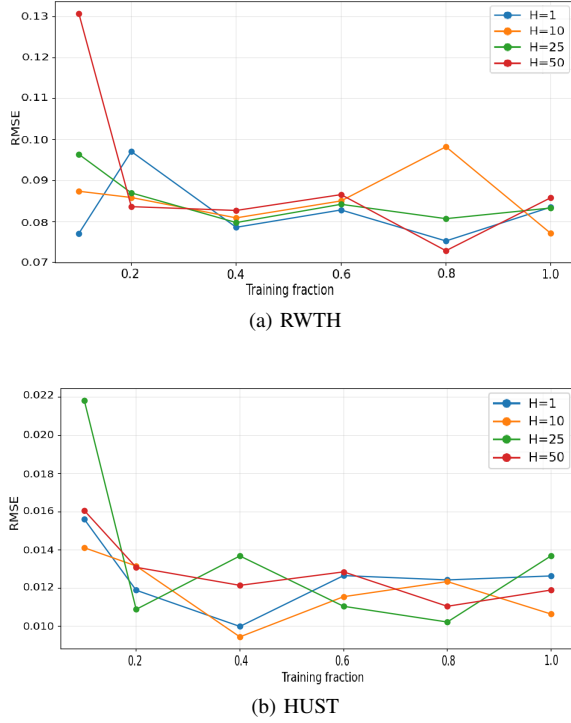


Fig. 6. Baseline learning curves (training fraction vs. RMSE) for RWTH and HUST across horizons $h \in \{1, 10, 25, 50\}$. Both saturate by 40–60% of the training data; for HUST the 40% fraction matches or beats full training at 42% lower training time.

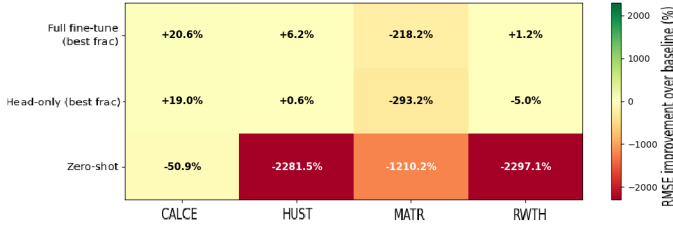


Fig. 7. Transfer RMSE improvement over baseline (XJTU source) across strategies and targets. The zero-shot row is uniformly catastrophic; fine-tuning turns CALCE, HUST and RWTH positive, while MATR remains negative under every strategy — the source-independent resistance seen from all sources.

It was further observed that head-only and full fine-tuning are close on the receptive targets at 40% and above, but head-only is markedly safer at 10% data, where full fine-tuning occasionally diverges; which is the catastrophic-forgetting risk the conference paper discusses. As well, the gap between the zero-shot bar and the fine-tuned bars is widest where domain shift is largest (RWTH, MATR), so zero-shot deployment is least reliable precisely in the cases where avoiding target-data collection would be most attractive.

XII. VERIFICATION AND VALIDATION

Several independent checks support the correctness of the results. Leakage control: splitting by battery ID, fitting the scaler on the training split only and constructing sliding win-

dows within a single cell’s history together ensure that reported accuracy reflects generalisation to unseen cells. Electrochemical plausibility: the empirical ranking of targets matches what battery chemistry predicts. The most transferable target (CALCE, LCO, a relatively predictable capacity fade) and the most resistant (MATR, fast-charge LFP, dominated by lithium-plating dynamics encoded in features absent from every other dataset) are exactly the targets electrochemistry would single out, which is strong corroboration that the model is responding to real degradation structure rather than artefacts. Baseline anchoring: source models were verified to reproduce their own test RMSE before being used for transfer, so a transfer failure can be attributed to the source–target relationship rather than to a defective source model. Negative controls: zero-shot transfer, which should fail if the domain shift is real, fails universally (51% to 3242% RMSE degradation; R^2 as low as -11.99 on RWTH, i.e. actively worse than predicting the mean), confirming that the improvements after fine-tuning are not an artefact of an easy problem. Reproducibility: a fixed seed, documented per-dataset preprocessing pipelines and saved forecasts and training histories allow each result to be regenerated.

These checks are reinforced by two diagnostic views of the predictions themselves, both collected in Annex A. Fig. 10 plots predicted against actual SOH for all three strategies on each target with XJTU as source. The zero-shot panels collapse to near-horizontal bands where the model emits an almost constant value irrespective of the true state, the behaviour that drives the R^2 of -11.99 on RWTH noted above. Full fine-tuning pulls the points onto the diagonal, recovering to $R^2 \approx 0.98$ on the same target. A transformation of this magnitude, achieved purely by adapting parameters on target data, is difficult to explain by anything other than genuine learning of the target’s degradation structure and is an assurance that the transfer data is sound. Fig. 11 shows the same behaviour in the time domain: overlaying each strategy’s prediction on the true SOH trajectory, the zero-shot curve is flat or offset, while the fine-tuned curves track the true degradation closely for RWTH, HUST and CALCE. MATR is again the diagnostic exception, even full fine-tuning fails to follow its fast-charge trajectory, which is consistent with the feature-content explanation for MATR’s resistance: the model cannot reconstruct dynamics encoded in features it never received. Read together, the scatter and overlay diagnostics confirm that the quantitative RMSE improvements correspond to qualitatively correct predictions, not to a reduced error norm.

A. Validation against literature benchmarks

External validation comes in two forms. The first concerns the baselines. Absolute RMSE values are not strictly comparable across papers since reported figures depend on the datasets, the preprocessing and whether SOH is normalised. A caveat that applies with particular force to RWTH’s Ah-scale capacity. Even so, the data-rich baselines obtained here sit in the same low-error regime reported in the literature for well-behaved datasets: HUST at $RMSE = 0.0095$ (0.95%)

and MATR at 0.0085 (0.85%) are comparable to the sub-1.44% RMSE regime that Huang et al. [4] report for their combined-health-indicator LSTM and are consistent in order of magnitude with the LSTM-based SOH and remaining-useful-life results of Zhang et al. [3] and Lu et al. [7]. This indicates that the from-scratch implementation is sound and that the weaker CALCE and XJTU baselines reflect those datasets’ data scarcity and intrinsic cycle-to-cycle noise rather than a modelling defect.

For the transfer findings, the universal zero-shot failure observed is consistent with the BatteryLife benchmark’s own report that models trained on one constituent dataset generalise poorly to others without adaptation [6]; the present study quantifies the severity (51%–3242% degradation) across all 20 pairs. The finding that fine-tuning reduces the target-data requirement is directionally consistent with Ren et al. [5], who showed fine-tuned models outperforming from-scratch baselines on a fraction of the target data; the divergence is in shape with their benefit accrues continuously with added data. Where here it activates as a step function at a discrete threshold, a refinement rather than a contradiction. The baselines match published accuracy levels and the transfer behaviour matches published cross-dataset trends, with this work adding the all-pairs, reliability-oriented characterisation those studies did not provide.

XIII. IMPLICATIONS FOR PRACTICE AND DEPLOYMENT

The reliability boundaries established in the conference paper translate into four recommendations for deploying LSTM-based SOH estimation across dataset boundaries; they constitute the practical outcome of the study.

First, always adapt before deployment. The universal zero-shot failure, 51% to 3242% RMSE degradation, with the model collapsing to a near-constant output shows that a source model must never be deployed on a new dataset without target-domain fine-tuning, regardless of how similar the chemistries or feature counts appear. In a safety-critical setting such as V2G/V2B, a zero-shot estimator that fails to register degradation at all is not merely inaccurate but dangerous, so this recommendation is a hard constraint rather than a preference.

Second, use head-only fine-tuning at the 40% fraction as the default. Across the receptive targets this configuration delivers accuracy within a small margin of full fine-tuning at roughly half the training cost and it avoids the catastrophic-forgetting risk that causes full fine-tuning to diverge when target data is scarce. The RWTH full fine-tune at 10% data, which inflates to $\text{RMSE} = 0.6974$, illustrates the risk. Freezing the LSTM backbone preserves the source’s temporal representations while the fully connected head adapts the output mapping, which is exactly the behaviour wanted when only a few target batteries are available.

Third, treat prediction-horizon alignment as a first-order selection criterion. Because horizon mismatch is a general failure mode that fine-tuning cannot undo, a practitioner choosing a source model should match its pre-training horizon to the target deployment horizon before considering chemistry or

feature overlap. A horizon-aligned source of modest feature overlap is a safer choice than a horizon-mismatched source of high overlap.

Fourth, collect target data only up to the activation threshold. Because transfer benefit activates as a step function and then plateaus, effort spent gathering target batteries beyond the threshold (around 40% of a typical training split in these experiments) yields negligible accuracy improvement at rising cost. The actionable form of this is to collect a small target set, fine-tune and stop once the threshold is crossed rather than pursuing a large target.

XIV. REPRODUCIBILITY AND THE RESEARCH PACKAGE

Reproducibility is essential to a reliability study, so the supporting research package is described here for completeness. Each dataset has a self-contained preprocessing pipeline that converts its native format — .mat for XJTU, .pkl for HUST, stitched Excel sheets for CALCE, zipped per-cell archives for RWTH, and JSON batches for MATR — into a common per-cycle feature table, so that every follow on experiment consumes data in an identical scheme. All randomness is controlled by a single fixed seed (42), the feature scaler is fitted on the training split alone, and the 70/15/15 split is taken by battery identity, so a rerun reproduces the reported numbers exactly rather than approximately. For every experiment the package retains the trained model, the per-cycle forecasts and the training history, which means any figure or table in the paper can be regenerated from saved artefacts without re-running the full sweep. The use of the BatteryLife benchmark as the data source adds a second layer of reproducibility: because BatteryLife standardises the raw cycling data and its splits, the datasets used here are the same as those any other researcher would obtain and the results are therefore comparable across studies rather than tied to a private preprocessing pipeline. A reproducibility gap is identified with the absence of multiple seeds, which means the reported values are exact point estimates but carry no uncertainty bounds. This is a gap the extension’s multi-seed study is designed to close.

XV. EXTENDED LIMITATIONS

The most consequential limitation is the single-seed design. Every reported number is a single-run point estimate; with one seed it is not possible to say whether small gaps for example, between head-only and full fine-tuning at 40% on CALCE are real or within the stochastic variation of weight initialisation. Quantifying this is the primary motivation for the extension. A second limitation is that the architecture is fixed throughout (hidden size 64, head [64-32-16-1]); the sensitivity of transfer to architectural choices such as hidden size or head depth is unstudied. Third, the feature-intersection alignment is deliberately simple and discards source information absent from the target; for MATR this reduces pre-training to head initialisation only, and more sophisticated alignment (shared-latent-space projection, adversarial alignment, similarity-guided source selection) might recover some

of the lost benefit. Fourth, each transfer uses a single source; multi-source pre-training is a natural and unexplored extension. Finally, two data limitations remain: a nine-batch MATR data set could not be incorporated because of an incompatible binary `.mat` format, restricting the fast-charge evaluation to three batches; and RWTH’s Ah-scale capacity would require explicit normalisation to be placed on the same footing as the other datasets.

Beyond these specific gaps, three threats to validity should be stated explicitly. External validity: the conclusions are drawn from five datasets and although the all-pairs design strengthens them relative to the usual two-to-four-pair studies, whether the two-regime structure and the activation threshold generalise to other chemistries and protocols is not yet established, it is exactly what the ten-dataset extension is meant to test. The construct validity for RWTH: because RWTH reports capacity in Ah rather than normalised SOH, its RMSE is not on the same scale as the other datasets, so RWTH results are read as within-dataset trends rather than cross-dataset magnitudes. The single-seed point-estimate risk already noted, which is most acute precisely where the reported differences are smallest: the narrow band of CALCE improvements across sources and the head-only-versus-full gap at the threshold are the two findings most in need of multi-seed confirmation, and the second-half plan prioritises them for that reason. Naming these threats does not weaken the conclusions; it bounds them and each maps onto a planned activity.

XVI. SYNTHESIS: FROM THE PRESENT FINDINGS TO THE EXTENSION

The five-dataset study produced four results that inform the second half. First, transfer benefit is real but conditional; it activates strongly for data-scarce targets and not at all for feature-unique ones. Second, the targets split into a source-independent regime (CALCE always improves, MATR always resists) and a source-dependent regime (HUST and RWTH, where source choice swings the outcome by tens of percentage points). Third, benefit activates as a step function at a discrete target-data threshold rather than accumulating smoothly. Fourth, prediction-horizon mismatch between pre-training and deployment is a general failure mode that fine-tuning cannot repair.

Each of these has a generalisation question that the current five-dataset, single-seed design cannot answer and it is those questions that the extension is built to close. Whether the two-regime structure and the activation threshold are properties of cross-dataset transfer in general, or artefacts of these particular five datasets, can only be tested by widening the benchmark hence Objective 1’s expansion to ten datasets. Whether the small but decision-relevant gaps; head-only versus full fine-tuning at the threshold, or the narrow band of CALCE improvements across sources are genuine effects or lie within run-to-run noise can only be settled by repeating the experiments across seeds and reporting standard deviations, hence Objective 2. The extension is not a new direction but the completion of the reliability argument the conference

paper begins: it converts qualitative, single-run boundaries into quantified, multi-dataset ones. Section XVIII sets out how this is to be done within the available time and compute.

XVII. CONCLUSION

At the mid-point of the project, several specific outcomes can be stated. A research gap has been established: the absence of a systematic, all-pairs evaluation of cross-dataset transfer for recurrent State-of-Health estimators spanning different chemistries, feature sets and dataset scales. From-scratch baselines have been built and characterised across five datasets, with the best window and horizon fixed per dataset and a computational-cost study showing that those baselines are already data-efficient. The full all-pairs transfer study has been completed; 20 source–target pairs and 140 experiments and has produced four solid results: zero-shot transfer is never reliable; fine-tuning compensates for target data scarcity, most clearly on the six-battery CALCE target; transferability splits into source-independent and source-dependent targets; and a prediction-horizon mismatch between pre-training and deployment is a failure mode that adaptation does not fix. Together these establish, for the datasets studied, when transfer across battery datasets can be relied upon and when it cannot.

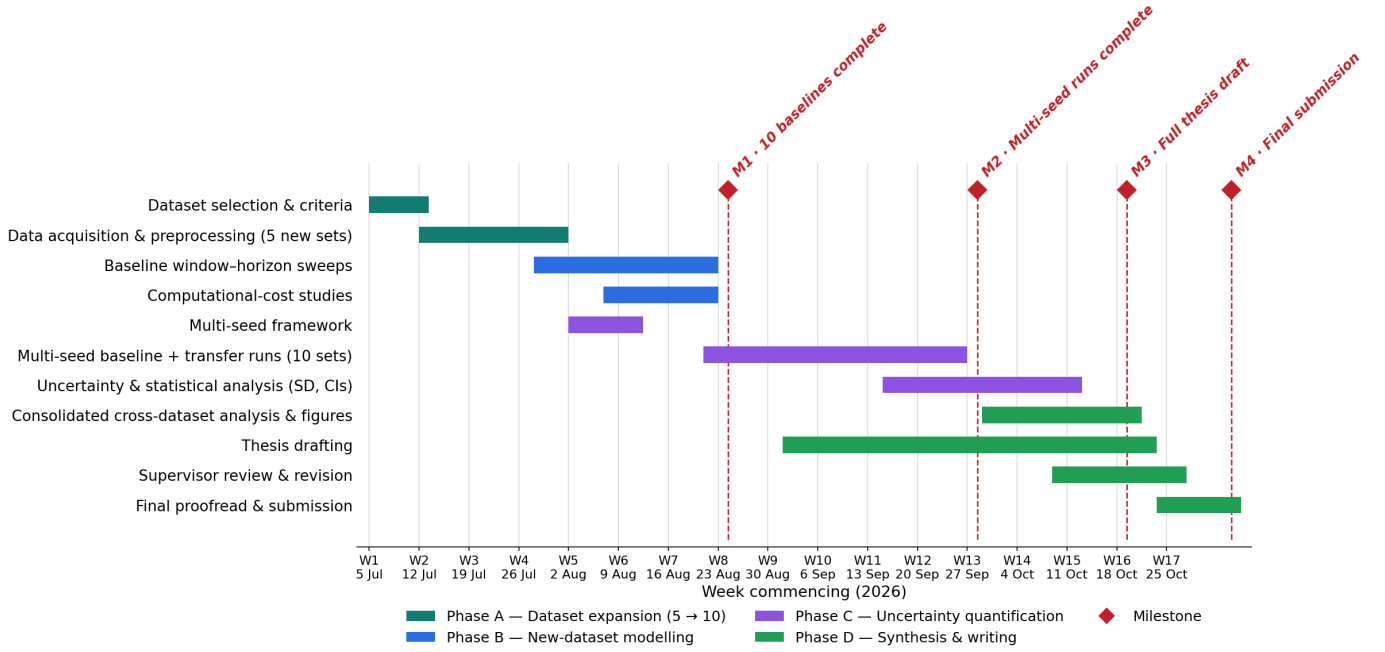


Fig. 8. Weekly Gantt chart for the extension phase (week commencing 5 July to end of October). Bars are grouped by phase (dataset expansion, new-dataset modelling, uncertainty quantification, and synthesis/writing); red diamonds mark the four project milestones.

XVIII. PLANS MOVING FORWARD

This section sets out the proposed second semester plan. It comprises two objectives doubling the benchmark from five to ten datasets and quantifying result uncertainty through multi-seed runs together with the computational strategy, risk treatment, and the weekly schedule shown in Fig. 8. The Gantt chart spans the week commencing 5 July to the end of October and is discussed in detail in the schedule subsection below.

A. Scope position

As established in Section V, the contracted deliverables are complete. The second semester is therefore an extension proposed on the strength of the results already obtained, not a recovery of unmet scope. The two limitations with the greatest bearing on the credibility of the findings; the narrow benchmark of five datasets and the single-seed point estimates. These map directly onto the two extension objectives below. If the extension could not be completed for any reason, the project as it stands would still satisfy its agreed deliverables; the extension raises the work from a complete five-dataset study to a broader and statistically quantified one.

The schedule and workload set out in this section assume that the requested extension is granted. The multi-seed objective in particular is compute-intensive: repeating the all-pairs design across ten datasets over several random seeds requires extensive unattended CPU time (Table III). If the extension is not granted, the achievable output within the standard timeframe would be substantially smaller, and the multi-seed uncertainty quantification would necessarily be limited to a prioritised subset of experiments rather than the full ten-

dataset grid; the schedule below is therefore predicated on the additional time the extension provides.

B. Objective 1 — dataset expansion from five to ten

The current study uses five BatteryLife datasets. The first objective is to double this to ten by incorporating five further BatteryLife constituent datasets as additional sources and targets. This widens the chemistry and operating-regime coverage and, more importantly, strengthens every cross-dataset claim: the two-regime structure, the activation threshold, and the horizon-mismatch failure mode are currently demonstrated on a five-dataset all-pairs design, and extending to ten tests whether they generalise. The work for each new dataset follows the established pipeline of format-specific preprocessing into the unified per-cycle feature table, a window-horizon baseline sweep to fix the optimal configuration, and a computational-cost study so the methodology is settled and the effort is principally one of careful, per-format data engineering and compute time rather than new design.

C. Objective 2 — multi-seed uncertainty quantification

The second objective is to repeat the experiments across the ten datasets over multiple random seeds, so that every reported RMSE and MAE carries a mean and a standard deviation rather than being a single point. This directly addresses the single-seed limitation in Section XV: it allows confidence intervals on the principal improvements and a principled answer to whether close results (such as head-only versus full fine-tuning at the activation threshold) are distinguishable. The deliverable is an uncertainty-annotated version of the main

TABLE III
APPROXIMATE EXPERIMENTAL SCALE OF THE EXTENSION RELATIVE TO
THE CURRENT STUDY.

	Current	Extension
Datasets	5	10
Baseline configs ($\times 20$ per dataset)	100	200
All-pairs source–target pairs	20	up to 90
Seeds per experiment	1	multi (~ 5)

results tables and the improvement heatmaps, with error bars on the data-fraction curves.

D. Experimental scale and computational strategy

The extension is computationally heavy, and its scale is stated explicitly to support realistic planning. Table III sets out the approximate growth in experiment count. The combination of ten datasets, an all-pairs transfer design, and multiple seeds is large, and on CPU it is the binding constraint on the schedule.

The strategy to keep this tractable is threefold: (i) reuse the pre-optimised window–horizon configuration per dataset so the seed sweep does not re-run the baseline search; (ii) prioritise the multi-seed effort on the experiments that carry the principal claims (the activation threshold, the two-regime targets and the horizon-mismatch cases) before broadening to the full ten-dataset all-pairs grid; and (iii) run the long seed batches as unattended background jobs, scheduled across the weeks shown in the Gantt chart. If full ten-dataset all-pairs multi-seed coverage proves infeasible within the schedule, the fallback is to report multi-seed uncertainty on the existing five-dataset all-pairs design and to use the five new datasets as additional single-seed targets, preserving both objectives in a reduced but defensible form.

E. Risks and contingencies

The principal risk is compute time: the multi-seed runs dominate the schedule and any slippage there compresses the write-up. This is mitigated by the prioritised ordering above and by the early start on data engineering, so that compute can run in the background while other work proceeds. A second risk is data-format incompatibility in the new datasets; the kind that blocked the nine-batch MATR data, this is mitigated by selecting the five new datasets early (Weeks 1–2) and validating their formats before committing to them. A third risk is that the new datasets do not all preprocess cleanly within budget; the contingency is to fix the number of usable new datasets at whatever is reached by the Objective-1 milestone and proceed to the multi-seed study on that set, rather than letting data engineering overrun into the analysis and writing phases.

F. Schedule

Fig. 8 shows the weekly schedule from the week commencing 5 July through the end of October, including the

write-up. The four milestones are: ten baselines complete (end of Week 7), multi-seed runs complete (end of Week 12), leaving time for a full thesis draft and final submission. Thesis drafting deliberately begins in Week 9 and overlaps the later experimental work, so that the settled methodology and the completed five-dataset results can be written up while the extension experiments run, leaving the final weeks for incorporating the multi-seed results, supervisor review, and proofreading.

APPENDIX

This annex collects the three full-width diagnostic figures referenced in Sections XI and XII: the all-strategy RMSE bar chart (Fig. 9), the predicted-versus-actual scatter grid (Fig. 10), and the SOH trajectory overlays (Fig. 11), all for XJTU as source. They are placed here to keep the body text uncluttered while preserving the full-resolution evidence behind the verification discussion.

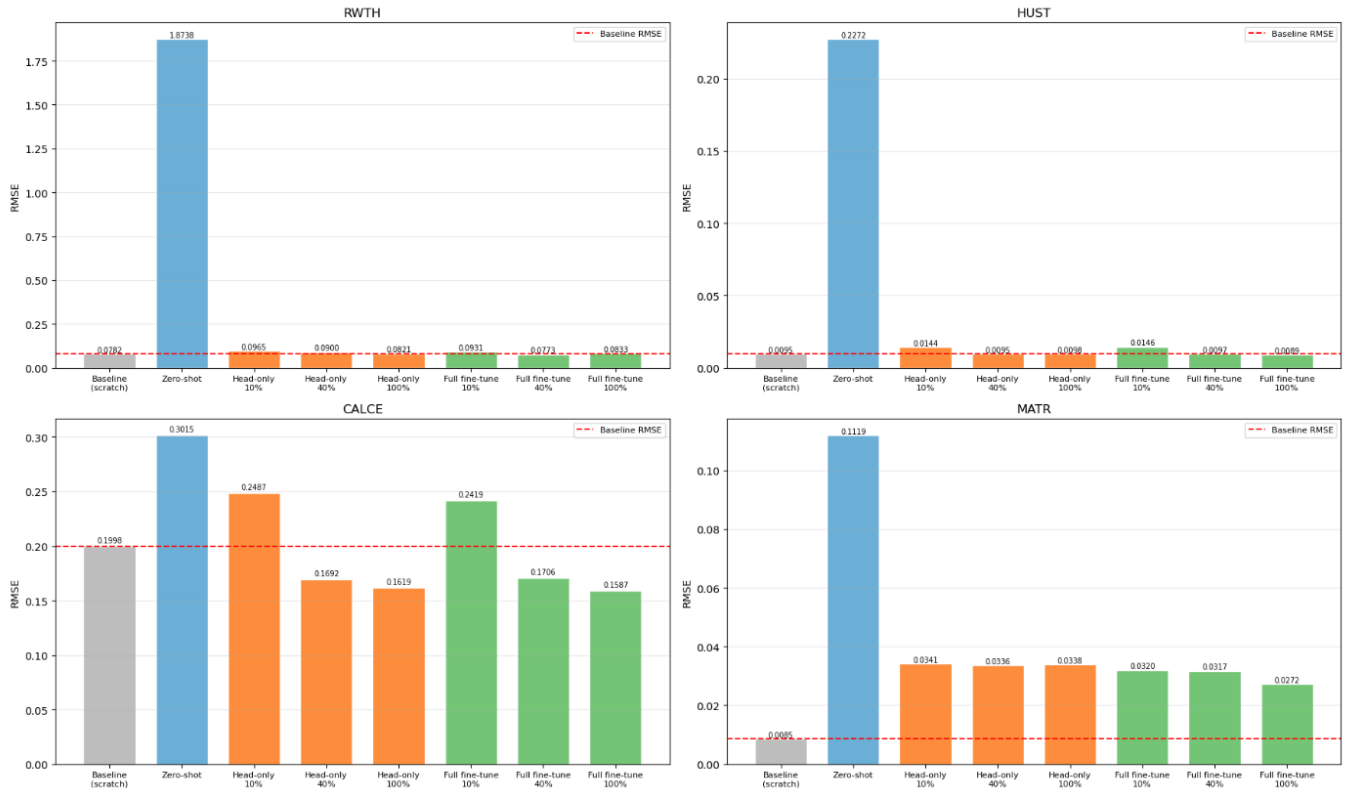


Fig. 9. RMSE for all seven transfer configurations per target (XJTU source), against the from-scratch baseline (dashed). Zero-shot towers above baseline on every target; fine-tuned bars fall to near or below baseline on CALCE, HUST and RWTH, while every MATR bar stays above it.

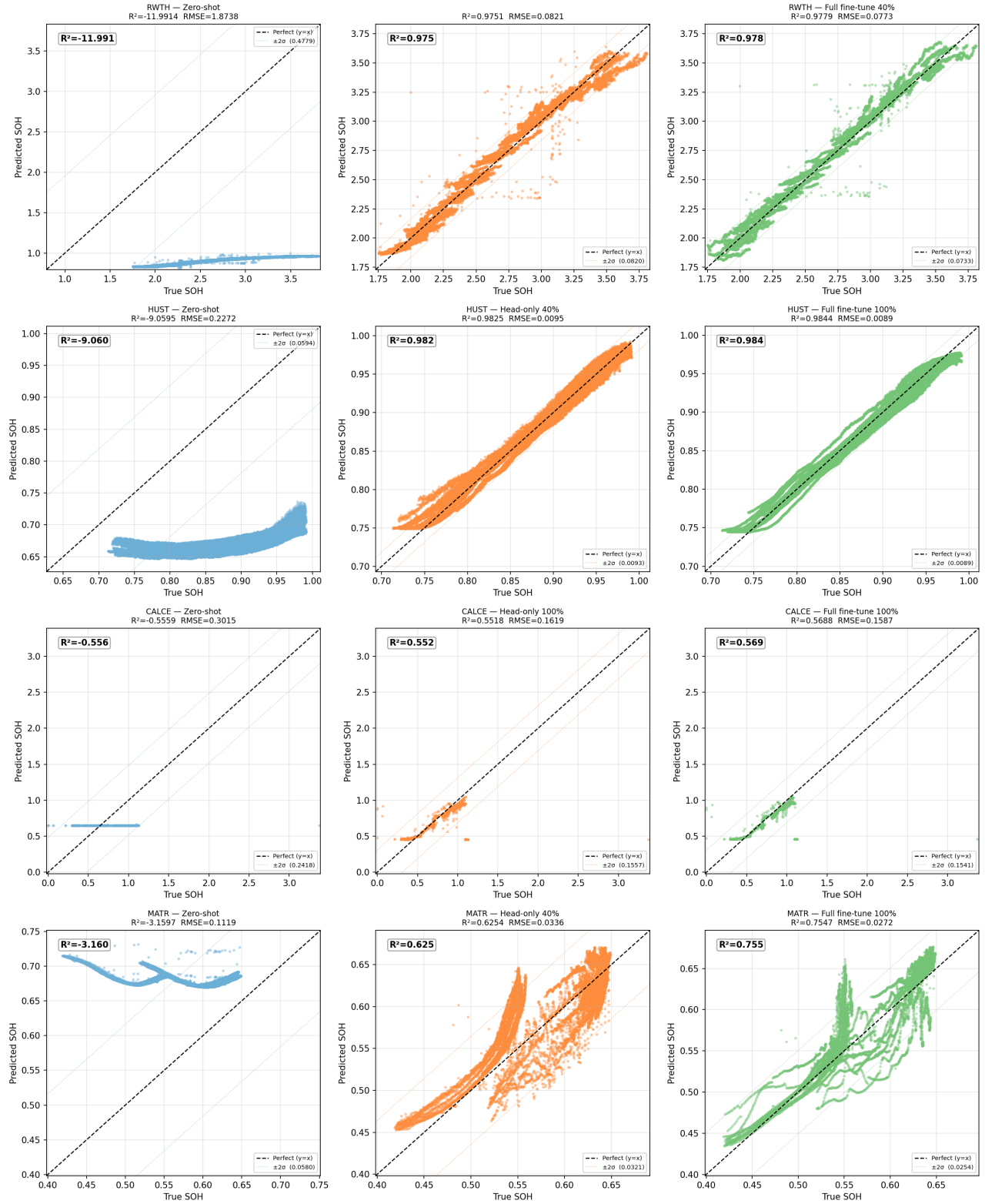


Fig. 10. Predicted vs. actual SOH (XJTU source). Rows: RWTH, HUST, CALCE, MATR; columns: zero-shot, head-only, full fine-tune. Zero-shot predictions form horizontal bands (near-constant output); fine-tuned predictions cluster on the diagonal, with full fine-tuning reaching $R^2 \approx 0.98$ on RWTH versus -11.99 zero-shot.

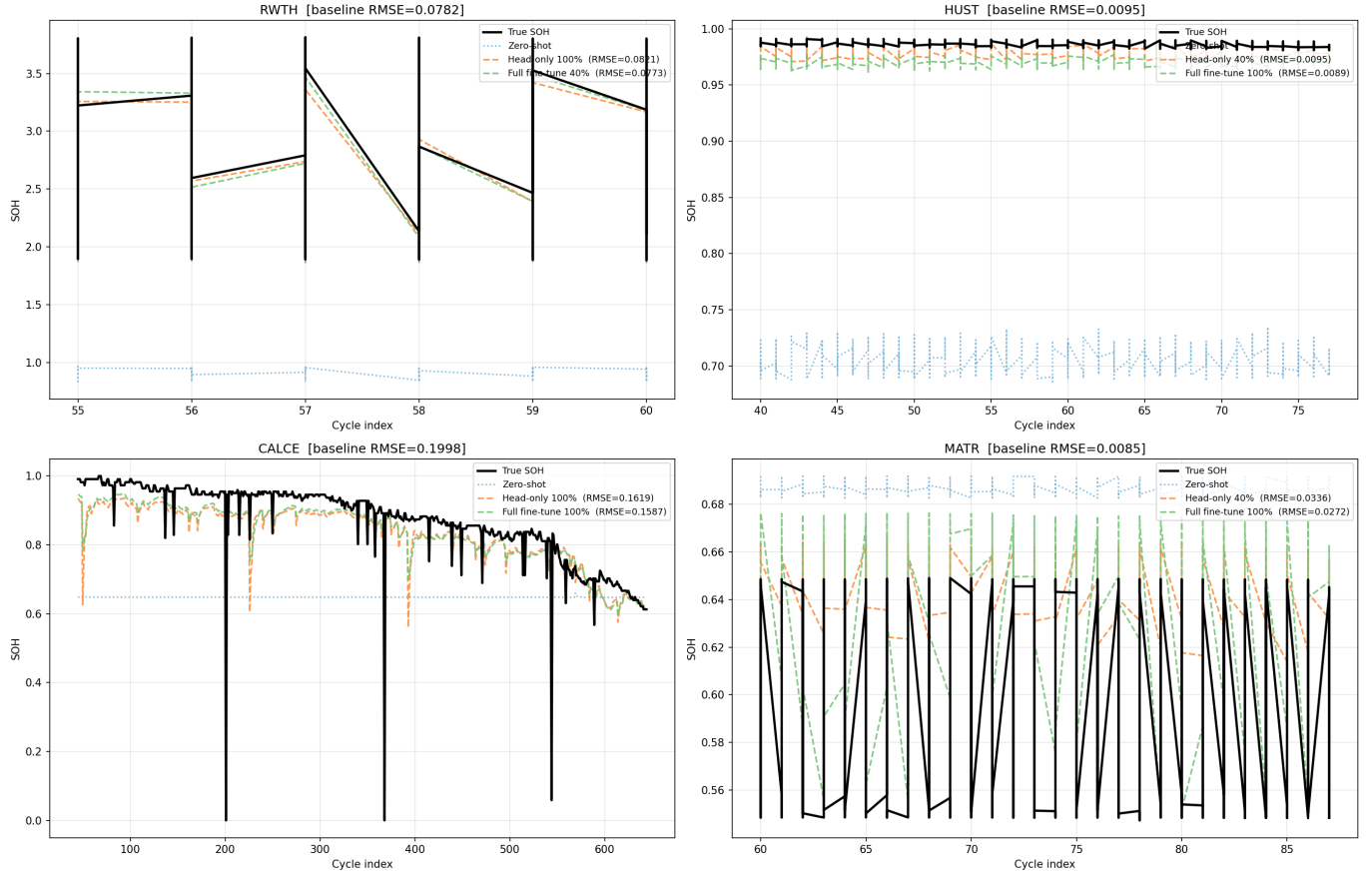


Fig. 11. SOH trajectory overlays (XJTU source). True SOH (black) against zero-shot (dotted), head-only (dashed) and full fine-tune (solid). Fine-tuned predictions track RWTH, HUST and CALCE closely; MATR remains poorly tracked even after full fine-tuning, consistent with its feature-content mismatch.

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